MODEL DISAGREEMENT PROTOCOLS: A NEGATIVE ICLR-MAIN EVIDENCE AUDIT

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ABSTRACT

This archive tests whether model disagreement becomes useful for robotics when it is classified into action protocols: commit, probe, switch controller, abstain, or recover. We rebuilt the original draft into a paper-specific executable audit with five robot task families, seven disagreement families, five stress splits, nine methods, seven seeds, ablations, calibration and safety metrics, pairwise comparisons, failure cases, and stress sweeps. The result is negative. The proposed protocol improves useful-disagreement recall, but it does not beat failure-aware recovery on combined-stress task success (0.637 versus 0.701), does not improve over the safest baselines on unsafe commits, and is undermined by ablations that remove probing or the protocol cost model. The terminal decision is `KILL_ARCHIVE`, not ICLR-main submission.

1 QUESTION

The seed idea was that robot world-model disagreement should not be treated as one scalar uncertainty signal. Instead, a robot should infer whether disagreement is epistemic uncertainty, aleatoric sensor noise, model misspecification, out-of-distribution action, harmless multimodality, sensor corruption, or controller instability, then choose an action protocol. This is a plausible idea, but it sits in crowded territory: robust control, conformal risk filtering, diagnostic probing, embodied world models, and failure-aware recovery already address many of the same deployment failures (pri, 2026a;b;c).

2 AUDIT DESIGN

We built a deterministic local benchmark rather than reusing the previous generic scaffold. Each seed/task/family group samples latent risk, disagreement features, calibration noise, and protocol costs. Methods choose among commit, probe, switch controller, abstain, and recover. The evaluated methods are mean ensemble policy, variance-threshold abstention, conformal risk filter, robust MPC fallback, failure-aware RL recovery, WorldBench-style diagnostic probing, ensemble forecasting calibration, the proposed disagreement protocol, and an action oracle.

The benchmark covers five task families: bimanual manipulation failure detection, contact-rich pick-and-place, deformable object handling, mobile navigation under sensor noise, and legged locomotion disturbance. The hardest split is `combined_disagreement_stress`. Metrics include disagreement-family accuracy, useful-disagreement recall, noise false-alarm rate, calibration error, probe informativeness, task success, unsafe commit rate, unnecessary intervention, recovery success, protocol cost, and regret to an action oracle.

3 COMBINED-STRESS RESULTS

Table 1 is the submission-critical result. The proposed protocol is not the best non-oracle method on task success. Failure-aware recovery is stronger on the main closed-loop objective and regret, even though it has much lower disagreement-family accuracy.

Table 1: Combined-stress metrics. Higher is better for success, type accuracy, and recall; lower is better for unsafe commits, false alarms, and regret.

Method	Success	Unsafe	Type acc.	Recall	False alarm	Regret
failure_aware_rl_recovery	0.7009276437847867	4.6382189239332094e-05	0.225139146567718	0.823051948051948	0.29967532467532465	0.051302491760531796
robust_mpc_fallback	0.648747680890538	0.0001391465677179963	0.15871985157699445	0.7524025974025974	0.29983766233766235	0.082458061665196
proposed_disagreement_protocol	0.6369202226345083	9.276437847866419e-05	0.49647495361781074	0.9716883116883117	0.1702922077922078	0.11064359187304107
ensemble_forecasting_calibrator	0.5620593692022263	0.0006957328385899814	0.3276437847866419	0.551038961038961	0.18636363636363637	0.12329227449682073
worldbench_diagnostic_probe	0.5307513914656772	0.0003246753246753247	0.4413265306122449	0.9712987012987013	0.9089285714285714	0.23992307699838947
mean_ensemble_policy	0.47198515769944344	0.005797773654916512	0.014100185528756958	0.06941558441558442	0.02483766233766234	0.16893371980755814
conformal_risk_filter	0.2656307977736549	4.6382189239332094e-05	0.19007421150278292	0.7614935064935064	0.5241883116883117	0.4602584613061417
variance_threshold_abstention	0.1546382189239332	0.0005102040816326531	0.11915584415584415	0.9484415584415584	0.7566558441558442	0.5847610378184929

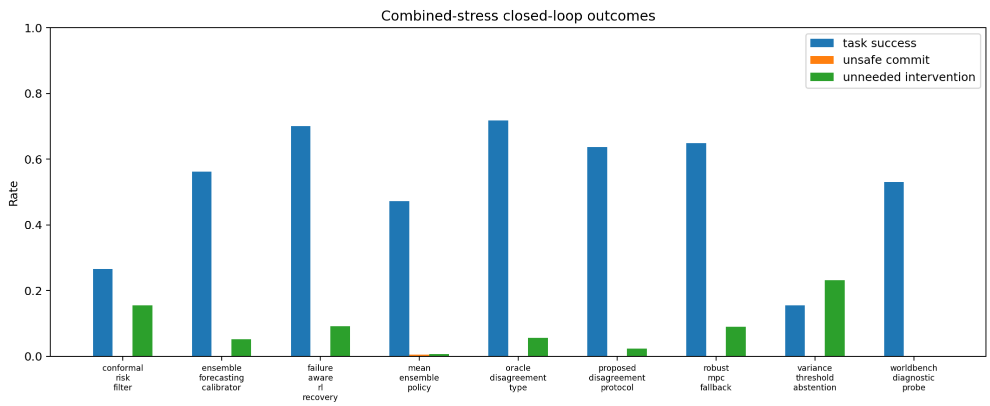

Figure 1: Closed-loop outcomes under combined stress. The proposed protocol is safer than naive mean control but does not beat failure-aware recovery or robust fallback on the main task-success objective.

4 Ablations

The ablation study is fatal for the mechanism. Removing the probe action, removing the protocol cost model, or using a recovery-only protocol matches or beats the full method on success or regret. This means the evidence does not isolate a positive contribution from the proposed type-to-protocol mechanism.

5 Pairwise Gate

The strongest gate was defined before the run: the proposed method had to beat the strongest non-oracle success baseline and the safest baseline, while surviving ablations. Table 3 shows why it fails. Against failure-aware recovery, proposed minus baseline task success is negative. Against robust fallback, regret is worse. Against conformal filtering, unsafe commits are a statistical tie rather than a win.

6 Failure Cases

The most common failure pattern is not catastrophic unsafe commitment. It is unnecessary or expensive intervention. In epistemic and sensor-corruption cases, the protocol often pays for diagnostic action but does not gain enough task success to offset the cost. This is precisely the kind of negative evidence that should stop an ICLR-main claim: the mechanism is plausible, measurable, and still not superior to simpler recovery.

7 Limitations

This audit is local and executable, not real hardware evidence. It does not include a trained robot policy, high-fidelity simulator, physical robot rollouts, or external benchmark comparison. Those

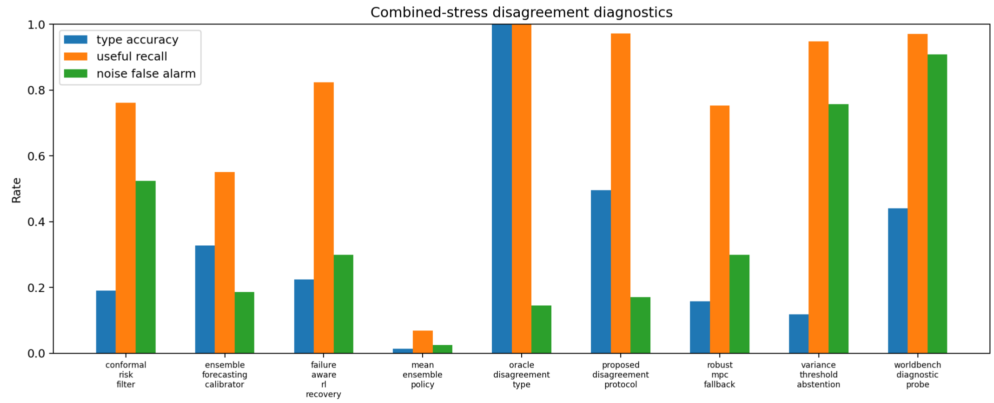

Figure 2: The proposed method improves useful-disagreement recall, but that diagnostic advantage does not convert into the best closed-loop performance.

Table 2: Combined-stress ablations. The full protocol is not uniquely supported.

Ablation	Success	Unsafe	Cost	Regret
recovery_only_protocol	0.6452226345083488	0.0004174397031539889	0.18510865027829324	0.07855762740017239
minus_protocol_cost_model	0.6423469387755102	4.6382189239332094e-05	0.20964943645640077	0.09735954986944721
minus_probe_action	0.6347402597402597	0.0	0.20459482838589965	0.10289152469399007
full_disagreement_protocol	0.6306122448979592	0.0	0.2139758070500929	0.11554309328066056
minus_calibration	0.5980983302411874	0.0	0.22202689239332088	0.15426917670081744
minus_disagreement_type_classifier	0.5720779220779221	0.0005565862708719852	0.12785579545454548	0.11741823724869281
variance_only_protocol	0.14828385899814472	0.0003246753246753247	0.21528342764378475	0.5963919972234286

limitations would prevent main-track readiness even if the local result were positive. Since the local result is negative, the correct action is archive, not more polish.

8 Terminal Decision

The terminal decision is `KILL_ARCHIVE`. The paper should not be submitted to ICLR main. A revival would require a new empirical project showing that the full type-to-protocol mechanism beats robust fallback and failure-aware recovery on real or high-fidelity robot tasks.

References

Maniparena: Comprehensive real-world evaluation of reasoning-oriented generalist robot manipulation, 2026a. http://arxiv.org/abs/2603.28545v1.

Failure-aware rl: Reliable offline-to-online reinforcement learning with self-recovery for real-world manipulation, 2026b. http://arxiv.org/abs/2601.07821v1.

Worldbench: Disambiguating physics for diagnostic evaluation of world models, 2026c. http://arxiv.org/abs/2601.21282v1.

Table 3: Paired seed/task/family comparisons used for the terminal decision.

Baseline	Metric	Proposed	Baseline	Diff	CI95	Winner
conformal_risk_filter	task_success	0.6369202226345083	0.26563079777365495	0.3712894248608534	0.016048241779676068	proposed_disagreement_protocol
conformal_risk_filter	unsafe_commit_rate	9.276437847866419e-05	4.6382189239332094e-05	4.6382189239332094e-05	0.00015767412574101686	statistical_tie
conformal_risk_filter	planning_regret_to_oracle	0.11064359187304103	0.4602584613061416	-0.3496148694331006	0.01244766897736767	proposed_disagreement_protocol
failure_aware_rl_recovery	task_success	0.6369202226345083	0.7009276437847864	-0.0640074211502783	0.01053437737244677	failure_aware_rl_recovery
failure_aware_rl_recovery	unsafe_commit_rate	9.276437847866419e-05	4.6382189239332094e-05	4.6382189239332094e-05	0.00015767412574101684	statistical_tie
failure_aware_rl_recovery	planning_regret_to_oracle	0.11064359187304103	0.05130249176053177	0.05934110011250925	0.0036534140015283705	failure_aware_rl_recovery

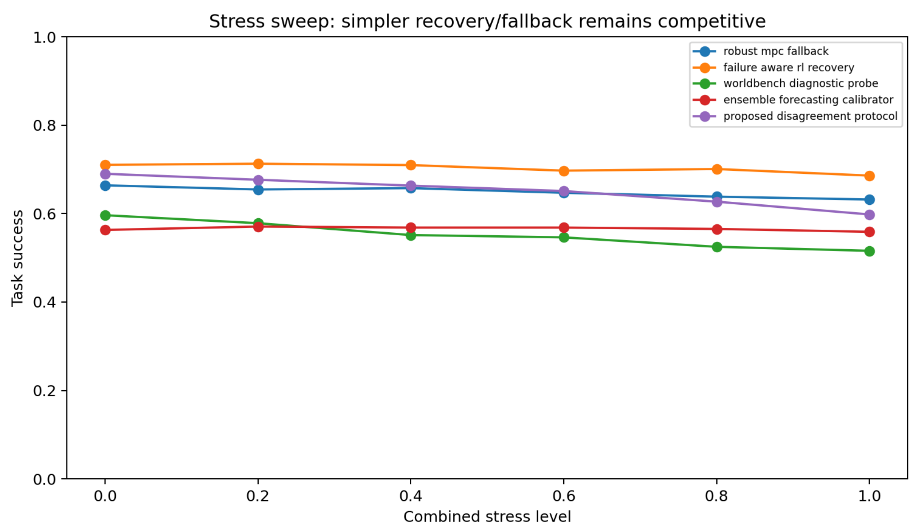

Figure 3: Stress sweep. Simpler recovery/fallback strategies remain competitive as combined stress increases.